

Neutral Inventions and the Stability of Growth Equilibrium*

1. Introduction

In criticising Hick's classification¹ of technical inventions, Harrod has proposed a new definition² of neutral inventions primarily intended for applications to the problem of economic growth. According to Harrod, a technical invention is defined as *neutral* if at a constant rate of interest it does not disturb the value of the capital coefficient. Harrod's classification has been discussed by J. Robinson [6] who showed graphically that a neutral invention is equivalent to "an all-round increase in the efficiency of labor" ([6], p. 140). The first part of the present article is concerned with precisely formulating Robinson's proposition and characterizing analytically those inventions that are neutral in Harrod's sense.

Harrod's definition of neutral inventions, as indicated above, has been introduced to handle the problem of economic growth. Recent contributions, however, in particular those of Solow [8] and Swan [9], are discussed for the case in which technical inventions are neutral in Hicks's sense.³ In the second part of this article we consider a neoclassical growth model with neutral inventions in Harrod's sense, and prove the stability of the growth equilibrium in such a model. The aggregate production function underlying the model is assumed only to be subject to constant returns to scale and to diminishing marginal rates of substitution; the Cobb-Douglas condition, as is customarily imposed in recent literature, is not required.

2. Neutral Inventions in Harrod's Sense

It is assumed that there are two factors of production: capital and labor. Both are assumed to be composed of homogeneous quantities; K and L refer respectively to the capital stock and the labor forces in terms of physical units. The structure of production in each time period t (conveniently referred to as *year* t) is described by the *aggregate production function* $F(K, L, t)$, which specifies the maximum output Y produced by capital K and labor L in year t :

$$(1) \quad Y = F(K, L, t).$$

Production is assumed to be subject to constant returns to scale; namely, the aggregate production function $F(K, L, t)$ is homogeneous of first degree:

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¹ According to Hick's definition, introduced in [3], pp. 121-2, a technical invention is termed *neutral* if the ratio of the marginal product of capital to that of labor remains undisturbed at a constant capital-labor ratio.

² Harrod's criterion was first introduced in his review of Robinson's book [1]; see Kaldor [4], Hicks [3], Robinson [5], pp. 132-6, Harrod [2], pp. 22-27, and Robinson [7], pp. 131-3.

³ Both Solow and Swan are concerned with the Cobb-Douglas production function case. The Cobb-Douglas production function is, as will be seen below, the only case in which Hicks's neutrality coincides with Harrod's neutrality. Hence, in Solow's and Swan's models, technical inventions are neutral in Harrod's sense also.

$$(2) \quad F(\lambda K, \lambda L; t) = \lambda F(K, L, t), \text{ for all } \lambda > 0.$$

Hence, the output per head, to be denoted by y , is uniquely determined by the capital-labor ratio, to be denoted by k :

$$(3) \quad y = f(k, t),$$

where

$$(4) \quad y = Y/L,$$

$$(5) \quad k = K/L,$$

and

$$(6) \quad f(k, t) = F(k, 1, t).$$

We assume that production is subject to the neoclassical conditions; namely, the aggregate production function $F(K, L, t)$ is continuously differentiable with respect to K, L, t ; capital is always productive, and the marginal rates of substitution are diminishing. The latter two conditions may be represented by

$$(7) \quad f_k(k, t) > 0,$$

and

$$(8) \quad f_{kk}(k, t) < 0,$$

where

$$f_k = \partial f / \partial k, \quad f_{kk} = \partial^2 f / \partial k^2.$$

The Cobb-Douglas function

$$F(K, L, t) = A(t) K^{\alpha(t)} L^{1-\alpha(t)}, \quad A(t) > 0, \quad 0 < \alpha(t) < 1,$$

is the most familiar neoclassical production function.

For a given rate of interest r , the optimum capital-labor ratio in year t , to be denoted by $k(r, t)$, is determined as the one at which the marginal product of capital equals the rate of interest r :

$$(9) \quad f_k[k(r, t), t] = r.$$

The corresponding optimum output per head $y(r, t)$ and capital-output ratio $x(r, t)$ are determined by

$$(10) \quad y(r, t) = f[k(r, t), t],$$

$$(11) \quad x(r, t) = \frac{k(r, t)}{y(r, t)}.$$

The technical invention represented by the aggregate production function $F(K, L, t)$, or equivalently by $f(k, t)$, will be termed *Harrod neutral* if the optimum capital-output ratio $x(r, t)$ is independent of t .

Under the diminishing marginal productivity assumption (8), the definition of the Harrod neutrality may be rephrased as:

$F(K, L, t)$ is Harrod neutral if and only if the marginal product of capital f_k remains undisturbed at a constant capital-output ratio x .

3. Robinson's Theorem on Neutral Inventions

An analytical characterization of the Harrod neutrality is given by the following:

Robinson's Theorem: The technical invention represented by $F(K, L; t)$ is Harrod neutral if and only if the production function $F(K, L; t)$ is of the form:

$$(12) \quad F(K, L; t) = G[K, A(t)L],$$

with a positive function $A(t)$.

Proof: By the assumptions (7) and (8), the relation (1) may be transformed into the one concerned with output per head y and capital-output ratio x :

$$(13) \quad y = \varphi(x, t)$$

where

$$(14) \quad k = xy.$$

The marginal product f_k of capital then can be expressed in terms of the function $\varphi(x, t)$ as follows:

Differentiating (11) and (12) with respect to x, y and k , we have

$$(15) \quad dy = \varphi_x dx,$$

$$(16) \quad dk = x dy + y dx.$$

Solving (15) and (16) with respect to dy and dk , we get

$$(17) \quad \frac{\partial y}{\partial k} = \frac{\varphi_x}{\varphi + x\varphi_x}.$$

Hence $f(k, t)$ is Harrod neutral if and only if the right-hand side of (17) is independent of t ; namely,

$$(18) \quad \frac{\varphi_x}{\varphi + x\varphi_x} = c(x),$$

where $c(x)$ is a function of x only.

From (18), we have

$$(19) \quad \frac{\varphi_x}{\varphi} = \frac{1}{\frac{1}{c(x)} - x}.$$

The relation (19) indicates that φ_x/φ is independent of t ; hence the function $\varphi(x, t)$ is decomposable:

$$(20) \quad \varphi(x, t) = A(t)\psi(x).$$

From (13) and (20), we have

$$(21) \quad x = \psi^{-1} \left[\frac{y}{A(t)} \right],$$

where ψ^{-1} is the inverse function of ψ .

The relation (21), together with (14), implies

$$(22) \quad \frac{k}{A(t)} = \frac{y}{A(t)} \psi^{-1} \left[\frac{y}{A(t)} \right];$$

hence,

$$(23) \quad \frac{y}{A(t)} = g \left[\frac{k}{A(t)} \right], \text{ say.}$$

In view of (3) and (4), the relation (23) may be written as

$$Y = G[K, A(t)L],$$

where

$$G(K, L) = g(k)L.$$

On the other hand, let the condition (12) be satisfied. Then, for the function $y = f(k, t)$, we have

$$(24) \quad y = f(k, t) = A(t) g \left[\frac{k}{A(t)} \right],$$

with $g(k) = G(K, L)/L = G(k)$.

Solving (24) with respect to y and x , we get

$$y = A(t) \psi(x)$$

for some $\psi(x)$. Hence

$$\frac{\varphi_x}{\varphi + x\varphi_x} = \frac{\psi'(x)}{\psi(x) + x\psi'(x)},$$

which is independent of t . Then, from (27), $\partial y / \partial k$ is independent of r ; hence $F(K, L, t)$ of the form (12) is Harrod neutral. Q.E.D.

It may be of some interest to compare Harrod's neutrality with Hicks's. According to Hicks [3], a technical invention is defined as *neutral* if the ratio of the marginal product of capital to that of labor remains unchanged at a constant capital-labor ratio. Such a technical invention may be termed *Hicks neutral*. It is possible to characterize the Hicks neutrality in terms of the aggregate production function $F(K, L, t)$; namely, the technical invention represented by $F(K, L, t)$ is Hicks neutral if and only if $F(K, L)$ is decomposable:¹

$$(25) \quad F(K, L, t) = A(t) F(K, L)$$

with some positive $A(t)$.

Corollary to Robinson's Theorem: The non-trivial technical invention represented by the aggregate production function $F(K, L, t)$ is both Harrod and Hicks neutral if and only if

$$(26) \quad F(K, L, t) = A(t) K^\beta L^{1-\beta},$$

with some positive $A(t)$ and $0 < \beta < 1$.

Proof: It suffices to show that the production function $y = f(k, t)$ is both Harrod and Hicks neutral if and only if

$$(27) \quad y = A(t) k^\beta, A(t) > 0, 0 < \alpha < 1.$$

Let $f(k, t)$ be Harrod neutral. Then

$$(28) \quad f(k, t) = B(t) g \left[\frac{k}{B(t)} \right],$$

with positive $B(t)$.

¹ This proposition is well-known; for an analytical proof, see, e.g., Uzawa and Watanabe [10].

On the other hand, the function $f(k, t)$ is Hicks neutral if and only if

$$(29) \quad \frac{\partial^2 \log f(k, t)}{\partial k \partial t} = 0, \text{ for all } k \text{ and } t.$$

Let the function $\psi(z)$ be defined by

$$(30) \quad \psi(z) = \log g(ez);$$

hence,

$$(31) \quad \log f(k, t) = \log B(t) + \psi\left(\log \frac{k}{B(t)}\right).$$

Differentiating (31) with respect to k and t , we get

$$(32) \quad \frac{\partial^2 \log f(k, t)}{\partial k \partial t} = \frac{-B'(t)}{k B(t)} \psi''\left[\log \frac{k}{B(t)}\right].$$

From (32) and (29),

$$\psi''\left[\log \frac{k}{B(t)}\right] = 0, \text{ for all } k \text{ and } t.$$

Therefore,

$$\psi(z) = \alpha + \beta z;$$

hence

$$g(k) = e^\alpha k^\beta.$$

Q.E.D.

4. Stability of the Neoclassical Growth Equilibrium

In this section we shall show the stability of growth equilibrium in a competitive model of economic growth with neutral technical progress in Harrod's sense.

Among the many recent contributions to the neoclassical theory of economic growth, the most important would probably be the one by Solow [8]; our formulation of a neoclassical growth model here is primarily based on Solow's.

We are concerned with an economy in which a homogeneous output is produced by capital and labor; any part of the output is either consumed by labor or accumulated as capital stock. For the sake of simplicity, it is assumed that capital never depreciates.

Let $K(t)$ and $L(t)$ be the capital stock and the labor force, respectively, and $Y(t)$ the rate of output, all in year t . The annual output $Y(t)$ is uniquely determined by capital stock $K(t)$ and labor force $L(t)$ in terms of the aggregate production function $F(K, L, t)$. It is assumed that the technical invention represented by $F(K, L, t)$ is always Harrod neutral; hence, by Robinson's theorem, the aggregate production function is of the form $F[K, A(t)L]$, where $A(t)$ may be referred to as the *efficiency of labor*. The function $F(K, L)$ satisfies all the neoclassical conditions.

In the neoclassical theory, both capital and labor markets are assumed *perfectly competitive*; hence the distribution of the annual output $Y(t)$ between capital and labor is determined by the marginal products of capital and labor. The return to capital $r(t)$ and the wage $w(t)$, in year t , are given by

$$(33) \quad r(t) = F_K[K(t), A(t)L(t)]$$

and

$$(34) \quad w(t) = F_L[K(t), A(t) L(t)].$$

We have

$$(35) \quad Y(t) = P(t) + W(t),$$

where $P(t)$ and $W(t)$ are respectively the shares of capital and labor in year t :

$$(36) \quad P(t) = r(t) K(t),$$

$$(37) \quad W(t) = w(t) L(t).$$

It is assumed for the sake of simplicity that labor does not save and capital does not consume. Then the neoclassical growth process may be described by the following system of differential equations:

$$(*) \quad \begin{cases} Y(t) = F[K(t), A(t) L(t)], \\ \frac{\dot{K}(t)}{K(t)} = F[K(t), A(t) L(t)], \end{cases}$$

where

$$\dot{K}(t) = \frac{dK(t)}{dt}.$$

It is assumed that both the rate of growth in labor, $\dot{L}/L$, and the rate of increase in the efficiency of labor, $\dot{A}/A$, are exogenously given and constant:

$$(38) \quad \frac{\dot{L}(t)}{L(t)} = \lambda > 0,$$

$$(39) \quad \frac{\dot{A}(t)}{A(t)} = \mu > 0.$$

Let $y(t)$ and $k(t)$ be respectively the output per head and the capital-labor ratio in year t :

$$(40) \quad y(t) = Y(t)/L(t),$$

$$(41) \quad k(t) = K(t)/L(t).$$

The growth process may be transformed into the one with respect to $y(t)$ and $k(t)$:

$$(**) \quad \begin{cases} \frac{y(t)}{A(t)} = f\left[\frac{k(t)}{A(t)}\right], \\ \frac{k(t)}{k(t)} = f_k\left[\frac{k(t)}{A(t)}\right] - \lambda. \end{cases}$$

Let

$$z(t) = \frac{k(t)}{A(t)}.$$

Then

$$(42) \quad \frac{\dot{z}(t)}{z(t)} = f_k[z(t)] - \lambda - \mu.$$

Since

$$f_{kk}(z) < 0, \text{ for all } z > 0,$$

the differential equation (42) is globally stable, and the equilibrium z^* is uniquely determined by

$$(43) \quad f_k(z^*) = \lambda + \mu.$$

Therefore, for the solution $[y(t), k(t)]$ of the process (**), both $y(t)/A(t)$ and $k(t)/A(t)$ converge:

$$\lim_{t \rightarrow \infty} y(t)/A(t) = y^*, \text{ say,}$$

and

$$\lim_{t \rightarrow \infty} k(t)/A(t) = z^*.$$

Hence,

$$\lim_{t \rightarrow \infty} x(t) = x^*,$$

where

$$x(t) = k(t)/y(t), \quad x^* = z^*/y^*.$$

The above observations lead us to the following:²

Equilibrium Theorem: Let the initial capital stock K^* and labor forces L^* satisfy

$$f_k[K^*/A(0)L^*] = \lambda + \mu,$$

where λ is the rate of growth in labor, defined by (38), and μ is the rate of growth in the efficiency of labor, defined by (39). Then, for the solution $[Y^*(t), K^*(t), L^*(t)]$ to the neoclassical growth process (*), the capital-output ratio $x^* = K^*(t)/Y^*(t)$ remains constant, and the output per head $y^*(t) = Y^*(t)/L^*(t)$ increases at the same constant rate as the capital-labor ratio $k^*(t) = K^*(t)/L^*(t)$. The capital-output ratio x^* is uniquely determined and may be referred to as the equilibrium capital-output ratio of the process (*).

*Stability Theorem:*³ Let the growth equilibrium exist. Then the neoclassical growth process (*) is globally stable; namely, for the solution $[Y(t), K(t), L(t)]$ to the process (*) with arbitrary initial $K(0)$ and $L(0)$, the capital-output ratio $x(t) = K(t)/Y(t)$ converges to the equilibrium capital-output ratio x^* .

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¹ If we measure the quantity of labor forces according to efficiency, then our stability theorem below is essentially reduced to the case without technical progress discussed by Solow [8].

² It has been pointed out by Professor Robert M. Solow that the stability of the neoclassical growth discussed here crucially hinges on the assumption that the balanced growth exists.

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